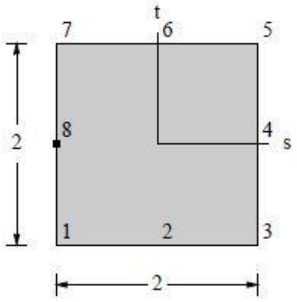
曲边四边形插值函数与映射单元（**mapped elements**）：

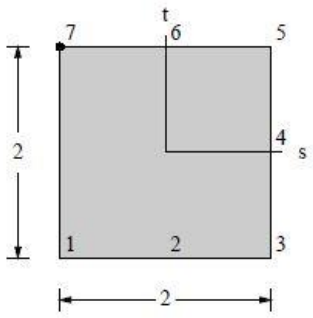
(i) Interpolation functions when all four sides are quadratic curves: 8 nodes

$$N=\begin{pmatrix} -\frac{1}{4}(-1+s)(-1+t)(1+s+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ \frac{1}{4}(-1+t)(1-s^2+t+st) \\ -\frac{1}{2}(1+s)(-1+t^2) \\ \frac{1}{4}(1+s)(1+t)(-1+s+t) \\ -\frac{1}{2}(-1+s^2)(1+t) \\ \frac{1}{4}(-1+s)(1+s-t)(1+t) \\ \frac{1}{2}(-1+s)(-1+t^2) \end{pmatrix}$$



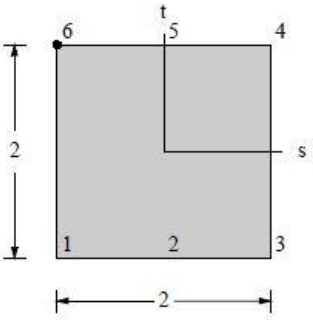
(ii) Interpolation functions when first three sides are quadratic curves: 7 nodes

$$N=\begin{pmatrix} -\frac{1}{4}(-1+s)s(-1+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ \frac{1}{4}(-1+t)(1-s^2+t+st) \\ -\frac{1}{2}(1+s)(-1+t^2) \\ \frac{1}{4}(1+s)(1+t)(-1+s+t) \\ -\frac{1}{2}(-1+s^2)(1+t) \\ \frac{1}{4}(-1+s)s(1+t) \end{pmatrix}$$



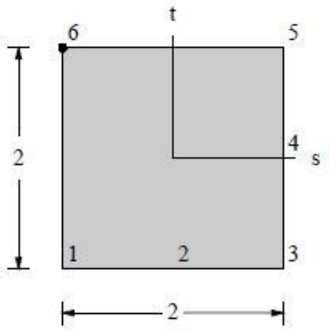
(iii) Interpolation functions when the opposite sides are quadratic curves: 6 nodes

$$N=\begin{pmatrix} \frac{1}{4}(-1+s)s(1-t) \\ -\frac{1}{2}(-1+s)(1+s)(1-t) \\ \frac{1}{4}s(1+s)(1-t) \\ \frac{1}{4}s(1+s)(1+t) \\ -\frac{1}{2}(-1+s)(1+s)(1+t) \\ \frac{1}{4}(-1+s)s(1+t) \end{pmatrix}$$



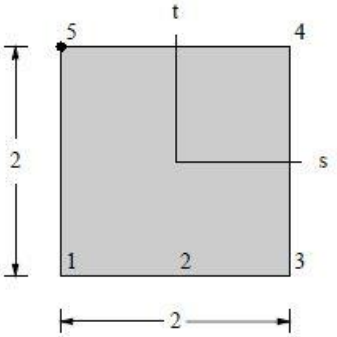
(iv) Interpolation functions when the two adjacent sides are quadratic curves: 6 nodes

$$N = \begin{pmatrix} -\frac{1}{4}(-1+s)s(-1+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ \frac{1}{4}(-1+t)(1-s^2+t+st) \\ -\frac{1}{2}(1+s)(-1+t^2) \\ \frac{1}{4}(1+s)t(1+t) \\ -\frac{1}{4}(-1+s)(1+t) \end{pmatrix}$$



(v) Interpolation functions when the first side is a quadratic curve: 5 nodes

$$N = \begin{pmatrix} -\frac{1}{4}(-1+s)s(-1+t) \\ \frac{1}{2}(-1+s^2)(-1+t) \\ -\frac{1}{4}s(1+s)(-1+t) \\ \frac{1}{4}(1+s)(1+t) \\ -\frac{1}{4}(-1+s)(1+t) \end{pmatrix}$$



Example 6.6 *Annular area*

Develop a mapping to a 2×2 square for the annular area shown in Figure 6.11. The inner radius is 2 units and the outer radius is 8 units. The circular arcs are defined by three points. The coordinates of all six key points are as follows.

	x	y
1	0	2
2	$\sqrt{2}$	$\sqrt{2}$
3	2	0
4	8	0
5	$4\sqrt{2}$	$4\sqrt{2}$
6	0	8

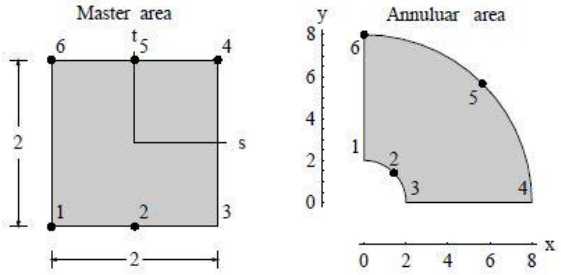


Figure 6.11. Six node master area and actual annular area

The node numbering is chosen in such a way that three nodes are placed on the first and the third side of the master element. The interpolation functions for this 6 node element are given in case (iii). Multiplying these interpolation functions by the x and y coordinates of the 6 key points of the annular area we get the following mapping.

$$x(s,t) = \frac{1}{4}(s-1)s(1-t)x_1 - \frac{1}{2}(s-1)(s+1)(1-t)x_2 + \frac{1}{4}s(s+1)(1-t)x_3 +$$
$$\frac{1}{4}s(s+1)(t+1)x_4 - \frac{1}{2}(s-1)(s+1)(t+1)x_5 + \frac{1}{4}(s-1)s(t+1)x_6 =$$
$$-\frac{3ts^2}{\sqrt{2}} + \frac{3ts^2}{2} - \frac{5s^2}{\sqrt{2}} + \frac{5s^2}{2} + \frac{3ts}{2} + \frac{5s}{2} + \frac{3t}{\sqrt{2}} + \frac{5}{\sqrt{2}}$$

$$y(s,t) = \frac{1}{4}(s-1)s(1-t)y_1 - \frac{1}{2}(s-1)(s+1)(1-t)y_2 + \frac{1}{4}s(s+1)(1-t)y_3 +$$
$$\frac{1}{4}s(s+1)(t+1)y_4 - \frac{1}{2}(s-1)(s+1)(t+1)y_5 + \frac{1}{4}(s-1)s(t+1)y_6 =$$
$$-\frac{3ts^2}{\sqrt{2}} + \frac{3ts^2}{2} - \frac{5s^2}{\sqrt{2}} + \frac{5s^2}{2} - \frac{3ts}{2} - \frac{5s}{2} + \frac{3t}{\sqrt{2}} + \frac{5}{\sqrt{2}}$$

The Jacobian matrix and the Jacobian of the mapping are as follows.

$$J = \begin{pmatrix} -3\sqrt{2}\,ts + 3ts - 5\sqrt{2}\,s + 5s + \frac{3t}{2} + \frac{5}{2} & -\frac{3s^2}{\sqrt{2}} + \frac{3s^2}{2} + \frac{3s}{2} + \frac{3}{\sqrt{2}} \\ -3\sqrt{2}\,ts + 3ts - 5\sqrt{2}\,s + 5s - \frac{3t}{2} - \frac{5}{2} & -\frac{3s^2}{\sqrt{2}} + \frac{3s^2}{2} - \frac{3s}{2} + \frac{3}{\sqrt{2}} \end{pmatrix}$$

$$\det J = \frac{9ts^2}{\sqrt{2}} - \frac{9ts^2}{2} + \frac{15s^2}{\sqrt{2}} - \frac{15s^2}{2} + \frac{9t}{\sqrt{2}} + \frac{15}{\sqrt{2}}$$

Setting detJ = 0 and solving for t we see that the Jacobian is zero when

$$t = -\frac{5}{3}$$

This is clearly outside of the master area and hence the mapping is good. The Figure 6.12 provides a graphically illustration of the mapping. A regular grid of points is chosen on the master area. For each pair of s, t values, the corresponding x, y values are computed from the mapping. All points are clearly mapped properly.

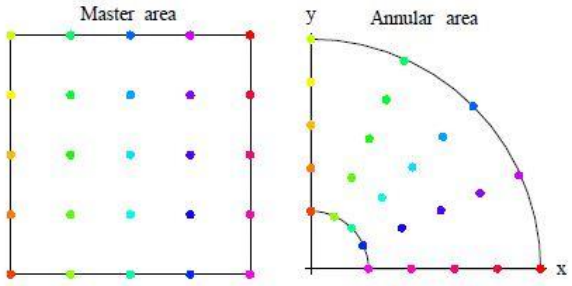


Figure 6.12. Mapping from the master area to the annular area

Example 6.7 *Quadrilateral with curved sides*

Develop a mapping to a 2×2 square for the quadrilateral area shown in Figure 6.13. All four sides are curved and are defined by three points on each side for a total of 8 points. The master element has eight nodes as shown in the figure. The coordinates of the key points are as follows.

	x	y
1	0.5	0.5
2	1.1	1.6
3	1.7	2.1
4	1.5	2.5
5	1.1	2.8

6	0.75	2.5
7	0.5	2.5
8	0.25	1.5

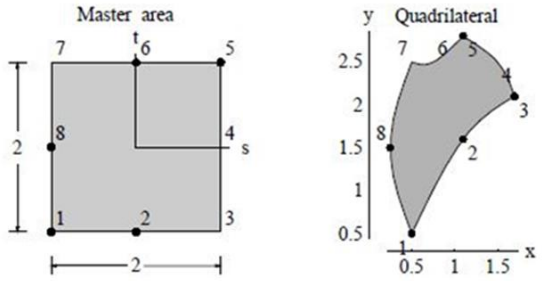


Figure 6.13. Eight node master area and actual quadrilateral

The interpolation functions for the master element are given in case (i). Multiplying these interpolation functions by the x and y coordinates of the 8 key points of the annular area we get the following mapping.

$$x(s,t) = -\frac{1}{4}(s-1)(t-1)(s+t+1)x_1 + \ldots + \frac{1}{2}(s-1)(t^2-1)x_8 =$$
$$0.025\,ts^2 + 0.025\,s^2 - 0.175\,t^2 s - 0.15\,ts + 0.625\,s + 0.075\,t^2 - 0.175\,t + 0.85$$

$$y(s,t) = -\frac{1}{4}(s-1)(t-1)(s+t+1)y_1 + \ldots + \frac{1}{2}(s-1)(t^2-1)y_8 =$$
$$0.225\,ts^2 - 0.075\,s^2 - 0.025\,t^2 s - 0.325\,ts + 0.5\,s - 0.025\,t^2 + 0.45\,t + 2.075$$

The Jacobian matrix and the Jacobian of the mapping are as follows.

$$J = \begin{pmatrix} -0.175\,t^2 + 0.05\,s\,t - 0.15\,t + 0.05\,s + 0.625 & 0.025\,s^2 - 0.35\,ts - 0.15\,s + 0.15\,t - 0.175 \\ -0.025\,t^2 + 0.45\,s\,t - 0.325\,t - 0.15\,s + 0.5 & 0.225\,s^2 - 0.05\,ts - 0.325\,s - 0.05\,t + 0.45 \end{pmatrix}$$
$$\det J = 0.015\,s^3 + 0.11625\,t^2 s^2 - 0.029375\,ts^2 + 0.089375\,s^2 -$$
$$0.123125\,t^2 s + 0.265\,ts - 0.131875\,s + 0.0125\,t^3 - 0.026875\,t^2 - 0.230625\,t + 0.36875$$

The Figure 6.14 shows the contour lines along which the Jacobian is zero. These lines are clearly outside of the master area and hence the mapping is good. The Figure 6.15 provides a graphically illustration of the mapping. A regular grid of points is chosen on the master area. For each pair of s, t values, the corresponding x, y values are computed from the mapping. All points are clearly mapped properly.

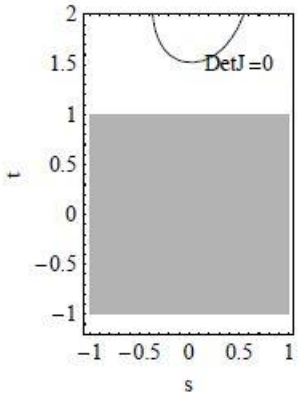


Figure 6.14. Contour for zero Jacobian

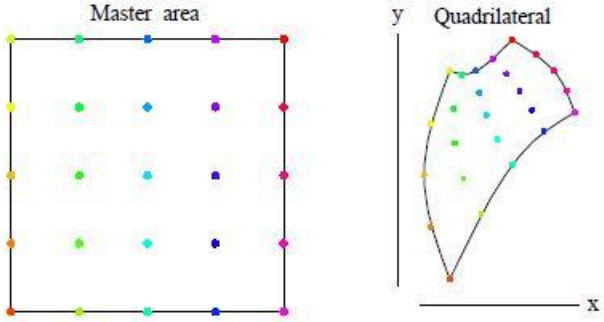


Figure 6.15. Mapping from the master area to the quadrilateral